



Predicting aggressive behavior in psychiatric patients in emergency department: A systematic literature review

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ABSTRACT

Introduction: Aggression and violence are major concerns in emergency departments (EDs), and have negative consequences for patient and staff health and safety. Few validated tools exist for identifying patients at risk of agitation. This study conducted a systematic literature review to identify and summarize the scores that predict aggressive behavior in EDs.

Methods: The search included articles published between Jan 1st, 1987, and Dec 31st, 2022, using the terms “aggress*,” “violent*,” “emergency,” “acute,” “score,” or “scale.”

Results: Ten scores were found to be relevant, with eight of the developed scores intended for use in EDs. The Aggressive Behavior Risk Assessment Tool (ABRAT) was found to be sensitive (84.3%) and specific (95.3%). The Brøset Violence Checklist (BVC) was highly specific (99.4%), whereas the Violence Screening Checklist (VSC) was less sensitive (57.2%) and specific (45.7%). The violence and aggression (OVA)/BVC checklist was found to significantly decrease the number of security call activations ($P < 0.001$). The Behavioral Activity Rating Scale (BARS) and OVA/BVC scores were the shortest, with seven and six items, respectively.

Conclusion: The OVA/BVC checklist is a valuable tool for predicting and preventing violence in the EDs. Future prospective studies should investigate its effectiveness.

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1. Background

Over the last decade, the incidence of mental illness has increased in the United States, while available services and funding are either stagnant or dwindling. This has created challenges in providing optimal care for Emergency Department (ED) patients with this pathology [1,2]. While funding shortages make it difficult to provide services to patients who seek care, the continued stigma surrounding mental health disorders prevents many from reaching out in the first place. The confluence of these trends has resulted in more patients with mental illness finding themselves in crisis and in need of emergent services [3].

The number of patients presenting to the emergency department with mental health complaints has been increasing, with an average of 774,508 mental health-related emergency department visits occurring annually among adults in the United States from 2018 to 2020, accounting for 12.3% of all ED visits by adults during that period. On average, 53.0 ED visits per 1000 adults occur each year in the United States,

where a mental health disorder is diagnosed, indicating a high demand for professional mental health services in the emergency department [4].

Aggressive behavior and agitation in psychiatric patients are significant challenges in the ED, where medical and traumatic emergencies are routinely managed and compete for limited time and space. The potential for escalation to aggressive behavior, which may put patients, staff, and providers at physical risk, makes it imperative to address agitated behavior rapidly and efficiently. Time constraints and limited access to psychiatric support have pushed emergency providers to frequently rely on chemical and physical restraints, a strategy called “restrain and sedate.” Recent research has demonstrated that a more patient-focused strategy is instrumental in enhancing the outcomes [5]. While agitation is a leading cause of hospital staff injuries and is associated with psychological suffering for both patients and medical staff, there are few unified approaches or standardized protocols to identify patients at risk of becoming agitated. Most commonly, institutions use intervention teams who arrive after an agitation event has begun and seek to engage patients in de-escalation and voluntarily accept treatment [6].

A standardized strategy to identify patients before agitation would allow this vulnerable population to be treated appropriately while

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increasing the safety of the medical staff. The potential for assessing the probability of a patient becoming agitated can be measured using objective risk scores, many of which are available. However, although many of the scores have a wide range of applicability, they may have limited utility in the ED setting. Thus, we performed a systematic literature review of aggressive behavior scores in psychiatric patients. Our goal was to summarize the existing agitation risk scores that predict aggressive behavior in psychiatric patients, determine their utility in the ED setting, and identify gaps in knowledge for future studies.

2. Methods

We performed a systematic search of PubMed, PsycINFO, and MEDLINE, databases for articles published from database from January 1, 1987, to June 30, 2023. We used the key words “aggression*,” “violence*,” “emergency,” “psychiatric,” “score,” and “scale,” using Boolean operators (AND, OR) for effective term combinations.

We screened abstracts for the following inclusion criteria: published in English, focus on psychiatric population and performed in an adult acute or emergency settings. We excluded studies published in language other English, case reports, editorials, conference papers, publications incorporating data from vulnerable populations (e.g., victims of domestic violence, prisoners), we also excluded publications focus only on one sex, or a specific diagnosis. We adhered to the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA).

Our Search strategy was meticulously documented, including the search terms used, the databases explored. This comprehensive approach aimed to ensure the rigor and thoroughness of our literature

review on the prediction of aggression in psychiatric patients within the emergency department setting. See Fig. 1.

3. Results

Of the 23 studies that met the inclusion criteria. These studies discussed 10 scores/scales. The included scores are presented in Table 1. The most common study methodology, occurring in seven studies of 46,993 patients, was prospective observational in nature, and two other studies performed retrospective evaluations.

The details of the individual studies are presented below.

3.1. Agitation Severity Scale (AgSS) [7]

An observation-based agitation scale for ED use in patients with an acute psychiatric presentation was evaluated and refined. They used a 21-item AgSS to assess 270 adult ED psychiatric patients in a prospective observational fashion. The relationship between AgSS and the previously established “Overt Agitation Severity Scale” was evaluated. The AgSS instrument was then reduced to 17 items representing the four categories of Aggressive Behaviors, Interpersonal Behaviors, Involuntary Motor Behaviors, and Physical Stance. AgSS validity was established by comparing its mean scores for subjects in the highest (most agitated) vs. the lowest (least agitated) scoring quartiles, with differences noted between these groups ($P < 0.001$), suggesting that the AgSS discriminated between subjects with and without high levels of agitation. The strengths of this validation study include its application in the ED and the identification of high-risk patients for agitation.

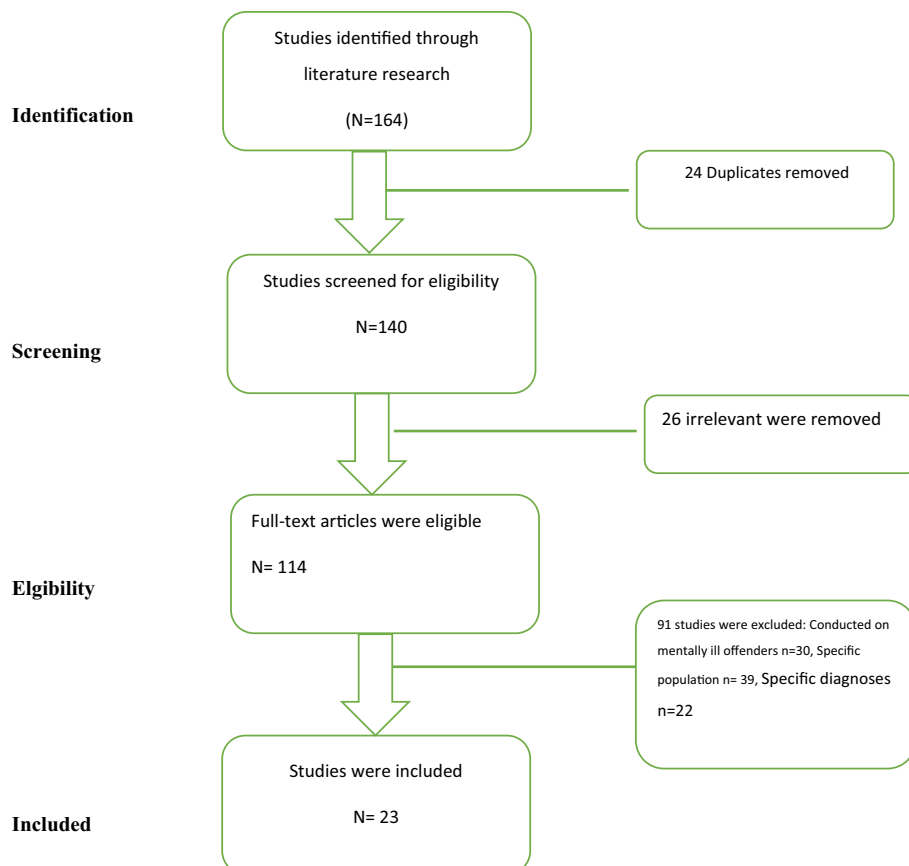


Fig. 1. Flowchart for the study selection process.

Table 1
The validation Studies of the aggression scores/scales evaluated in our investigation.

Score/Scale	Validation study & year of published	Settings & Study design	Results	Limitations
1. Agitation Severity Scale (AgSS)	Strout [7] (2014)	Emergency department; prospective observational study to assess the ability of the Agitation Severity Scale (AgSS) to identify patients with high risk of agitation	$N = 270$; the Agitation Severity Scale (AgSS) identified patients at high risk of agitation by comparing mean scores for subjects in the highest vs. the lowest scoring quartiles ($p < 0.001$)	- Population (primarily of white race), - Unblinded primary investigator;
2. Behavioral Activity Rating Scale (BARS)	Legambi et al. [21] (2021)	Urban ED, Quality Improvement project. Data was collected retrospectively, and nurses survey prospectively. The study aimed to help nurses to use Behavioral Activity Rating Scale (BARS) correctly to predict violence.	$N = 780$ charts reviewed. There was no difference in the number of restraints used. There was numerical trend for patients to spend less time in restraints.	- Numbers of aggressive episodes not clear. Did not reduce violent incidents, security activations or the use of mechanical restraints.
3. a. Brøset Violence Checklist (BVC)	Partridge B [7,16], Affleck (2018)	Metropolitan hospital ED in Australia. Prospective observational study to evaluate the ability of the Brøset Violence Checklist (BVC) to predict violence when used by officers in the triage area of the ED	$N = 2064$. 1.7% committed a violent/aggressive act. Brøset Violence Checklist (BVC) sensitivity was 45.7% and specificity 99.4%. At a cut-off score of 3, Positive Predictive Value (PPV) was 55.2%. Violent patients were around 71 times more likely to score BVC ≥ 3 than non-violent patients.	Brøset Violence Checklist (BVC) scored by security officers.
3. b. Occupational Violence and aggression (OVA) – implementation of BVC	Ainslie Senz et al. [17] (2021)	ED in Australia. A before-and-after implementation study evaluating violent episodes, and a survey study of nursing staff. evaluated the impact of using Occupational Violence and aggression in ED and the rate of security events related.	$n = 250$ There was a reduction in unplanned security responses (relative risk 0.75, 95% CI 0.62–0.89, $P = 0.001$). There was early detection and intervention (relative risk 2.22, 95% CI 1.85–2.66).	It was tested in a single healthcare system in Australia.
4. Dynamic Appraisal of Situational Aggression (DASA)	Ogloff and Daffern [18] (2006) Hamric et al. [20] (2023)	In-patients' psychiatric unit. Before and after observational study.	$n = 100$, Nurses accuracy in predicting aggression was higher after the score implementation ($P < 0.001$) with cutoff score of 4	Inpatient study
5. Positive and Negative Syndrome Scale Excited Component (PANSS-EC)	Montoya et al. [22] (2011)	Four non-psychiatric Eds. Retrospective chart review to examine the utility of the DASA tool to identify and reduce potential aggressive events by ED patients and coordinate DASA with appropriate interventions.	$n = 498$, increased prediction of violence incidents ($p = 0.02$). Score < 2 required no intervention. > 4 required escalation	It requires training and it was implemented in single system.
6. Clinical Global Impression Scale for Aggression (CGI-A)	Christain Huber [25] (2008)	Multicenter Eds. prospective observational study to evaluate the Positive and Negative Syndrome Scale Excited Component (PANSS-EC) as a tool to identify patients at risk of aggression.	$n = 278$ patients required oral medications. 8 experts assessed the score and considered it useful on scale of 5 (mean 3.3 SD 0.6)	The use of only observations parameter, lack of impact measures and exclude patients required parenteral medications.
7. Staff Observation Aggression (SOAS-RE)	Morken et al. [26] (2018)	In-patients' psychiatric unit. Prospective to assess Clinical Global Impression Scale for Aggression (CGI-A) as a screening tool of aggression. Ten ED in Norway. Prospective observational study.	$n = 558$, spearman correlation with Positive and Negative Syndrome Scale Excited Component (PANSS-EC) $r = 0.83$.	It is not ED settings and needs training and experience with the psychiatric patients.
8. Violence Screening Checklist (VSC)	Bjørkly [28,31] (2009)	The aim was to adjust a pre-existing form for reporting aggressive to the ED and assess the validity of the score. University short term in-patient's psychiatric unit. Prospective observational study to validate the checklist.	$n = 291$, it showed low correlation with Visual Acuity Scale (VAS; $r = 0.45$).	It was evaluated in one healthcare system, poor correlation, observation parameter and clinical outcomes measures were absent.
9. Aggressive Behavior Risk Assessment Tool (ABRAT)	Kim et al. [31] (2022)	Three EDS. Prospective observational study to validate the tool in the ED.	$n = 338$, score of > 2 identified high risk of violence. Positive Predictive Value (PPV) (59.0%), Negative Predictive Value (NPV) (70.6%), sensitivity (57.2%), specificity (70.0)	It was used in in-patients' settings only.
10. The emergent Documentation Aggression Rating Tool (EDART)	Campbell et al. [32] (2021)	ED of a midwestern suburban hospital. Quality improvement study to evaluate tool's feasibility, identify escalating behaviors and implement interventions.	$n = 10,554$ patients > 10 yrs. 127 violent incidents. Cutoff 1 sensitivity and specificity of 84.3% and 95.3%, respectively. with a Predictive Value (PPV) and Predictive Value (NPV) of 8% and 99.8%, respectively. 32,290 all ED visits were screened.it showed marked increase of historical violence potential documentation rates from 67.4% to 97.6% and a trend to decreased restraint use.	Violent incidents were only ($n = 127$), low compliance rate (44%), it was performed during the COVID-19 pandemic, which may effect is generalization It was applied on all ED patients not patients with psychiatric complaints only. Descriptive study showing improved documentation increases security activations.

However, limitations included a biased representation of the ED population (primarily of white race), enrollment on a convenience basis (sample bias), and data collection by an unblinded primary investigator.

3.2. Behavioral Activity Rating Scale (BARS)

The BARS [8] was developed by pharmaceutical companies to assess agitation in drug trials and divided patients into seven levels of agitation: unable to arouse, very sedated, sedated, calm, agitated, very agitated, and dangerous agitation. Swift et al. [9] compared the BARS to a previously standardized Clinical Global Impression (CGI; National Institute of Mental Health, 1976) [10] and Positive and Negative Syndrome Scale for schizophrenia (PANSS) (Kay et al., 1987) [11]. BARS showed a moderate correlation with PANSS and CGI-S ($P < 0.005$) and was able to differentiate between agitated and calm states in those with psychosis; hence, it was used to detect the efficacy of medications used to treat agitation [12]. In a pre- and post-implementation study, they found that although there were no differences in the rate of restraint use, there was a numerical trend for patients to spend less time on restraints. The strengths of these evaluations include the fact that BARS can objectively rate agitation and that it was tested in the ED. Consequently, it has been recommended by the “Best Practices for Evaluation and Treatment of Agitated Children and Adolescents” was used to assess agitated patients in the ED [13]. Its limitations include the lack of demonstrable ability to reduce violent incidents, security activations, or the use of mechanical restraints.

3.3. Brøset Violence Checklist (BVC)

The BVC [14] is a short-term violence prediction instrument that assesses confusion, irritability, boisterousness, verbal threats, physical threats, and attacks on objects, all rated as present or absent [15]. The BVC discriminates between violent and non-violent activities over the ensuing 24-h period. The BVC was applied in the ED, assessing 2064 patients, of whom 35 (1.7%) committed violent or aggressive acts. The sensitivity, specificity, and positive predictive value for predicting a violent or aggressive act were 45.7%, 99.4%, and 55.2%, respectively, at a cut-off score of >2 . Violent patients were approximately 71 times more likely to have a BVC ≥ 3 than were non-violent patients [16].

The BVC was further validated [17] by its inclusion in an occupational violence and aggression recognition program applied in an Australian ED. In a before and after study design, using a score-based notification and response framework, they found that the documentation of assessing violence improved from 30% to 82% ($P < 0.0001$), there was a reduction in emergency security responses (relative risk 0.75, 95% CI 0.62–0.89, $P = 0.001$; absolute risk reduction 0.18% (95% CI 0.07–0.29%) and that planned security responses increased (relative risk 2.22, 95% CI 1.85–2.66, $P < 0.001$), although there was no difference in the proportion of patients undergoing mechanical restraint. The overall strengths of these evaluations include ED application and clinically valuable endpoint changes following implementation, although limitations include the lack of comparison with previously validated tools.

3.4. Dynamic Appraisal of Situational Aggression (DASA)

The DASA [18] score is based on the Brøset-Violence-Checklist (BVC) [14] to assess the likelihood that a patient will become aggressive within the inpatient psychiatric environment. It scores 0 or 1, for absent or present in the last 24 h, for the following parameters: irritability, impulsivity, unwillingness to follow instructions, sensitivity to perceived provocation, easily angered when requests were denied, negative attitudes, and verbal threats. In the initial 100 patients, before and after inpatient psychiatric unit study of score implementation, nurses were asked to rate the likelihood of a patient becoming aggressive. Their accuracy in predicting aggression was higher after score implementation ($P < 0.001$), with a score cut-off of 4 and a C-statistic of 0.82. Kaunomäki

et al. [19], using a cut-off of 4 to identify high-risk individuals, found similar results and noted that the odds of violence for a patient scoring 7 was 29 times higher than that of those scoring 0.

Finally, a quasi-experimental study by Hamric et al. [20] enrolled 498 patients from four non-psychiatric EDs. The DASA was applied at triage and 8 h intervals during the ED encounter. They reported that although training was required before implementation, DASA was feasible in the ED and increased the prediction of violent incidences ($P = 0.02$). They also found that a score of <2 required no intervention, whereas scores >4 required escalation and physician intervention. The strength of the DASA is that it was evaluated in the ED, which was quick and easy to perform, and it increased the predictive accuracy of nurses in identifying patients with a high likelihood of violence. Its limitations are that it requires training to be implemented and that it has been evaluated only within a single system.

3.5. Positive and Negative Syndrome Scale Excited Components (PANSS-EC)

The PANSS-EC is a shorter version of and adapted from the original PANSS [21], a tool to assess the symptoms of schizophrenia and other general psychiatric diseases. The PANSS-EC consists of five items rated from 1 (absent) to 7 (extremely severe): excitement, tension, hostility, uncooperativeness, and poor impulse control. The total score ranges from 5 to 35, with a score of ≥ 20 clinically corresponding to severe agitation. In a prospective observational study, Montoya et al. [22] tested PANSS-EC in patients with acute psychosis and agitation. Enrolling 278 psychiatric patients who required oral medication per standard care, they applied PANSS-EC and two other well-known agitation severity scores (The Agitation and Calmness Evaluation Scale (ACES), which is a tool used to evaluate the severity of agitation in bipolar patients and schizophrenia [23] and the Clinical Global Impressions-Severity scale (CGI-S)) [24].

Scores were applied on ED arrival and at discharge or transfer to the psychiatric unit, but patients requiring IV medication were excluded. Eight experts were queried and it was found that the scale eased their assessment of agitation intensity in patients with acute psychotic episodes and during their follow-up. They considered PANSS-EC to be useful. The analysis of each item's importance, frequency, and clarity on a 5-point scale showed a mean value between 4 and 5 for most items, except for clarity in the tension, lack of cooperation, and excitement items, which showed a mean of 3.3 (sd 0.6). Although the strengths of this evaluation include its performance in an ED and its feasibility, its limitations include the use of only observational parameters, exclusion of patients requiring intravenous medications, and lack of clinical endpoints [22].

3.6. Clinical Global Impression Scale for Aggression (CGI-A)

The CGI-A score quantifies aggression on a scale from 1 to 5 (1 = no, 2 = slight, 3 = moderate, 4 = severe, and 5 = aggressive behavior present), and was initially used in patients with schizophrenia. This score was evaluated in 558 hospitalized psychiatric patients who were continuously assessed over 5 days. The authors compared the results obtained by the CGI-A with those of the Positive and Negative Syndrome Scale-Excited Component (PANSS-EC), a previously validated rating scale for agitation, excitement, and aggression in schizophrenia spectrum disorders. Spearman's correlation coefficient between the CGI-A and PANSS-EC total scores was $r = 0.83$. The strengths of this scale are that it is easy and quick to use; however, it is limited by the lack of data in the ED setting and the need for psychiatric staff for its administration [25].

3.7. Staff Observation Aggression Scale-Revised (SOAS-RE)

The SOAS-RE [26] is an updated version of the original SOAS [27], which is an incident report form used by nursing staff to monitor

aggression frequency, nature, and severity. In the new application, aggression is defined as damaging or threatening objects or humans and is evaluated in five categories (score range in parenthesis) to assess aggression severity, which includes:

1. The circumstances that provoked the aggressive behavior (provocation) (0–2).
2. The means used to act out the aggressive behavior (0–3).
3. The target of the aggression (0–4).
4. The effect of aggressive behavior on the environment (0–9).
5. The measures taken to stop the incident (0–4).

Thus, the summation range of the scale is from 0 to 22, with 0 to 7 considered mild, to 8–15 moderate, and > 16 severe aggression. This score was evaluated in a prospective observational study of 291 patients at 10 sites in Norway, comparing the SOAS-RE to a visual analog scale of the estimate of aggression. The mean (SD) SOAS-RE score was 10.0 (4.0) vs. a mean (SD) VAS of 45.4 (26.7), but with a low correlation (Pearson = 0.45, $P = 0.001$). Linear regression analysis showed that each observational category had a low impact on the VAS score. The strength of this score is that it was tested in the ED and does not require special training for its application. However, it has only been evaluated in Norway and has poor correlation with other evaluation methods. Individual evaluation parameters had a marginal impact and clinical outcome evaluations were absent.

3.8. Violence Screening Checklist (VSC)

VSC [28] was used to assess the risk of inpatient violence. The parameters were rated as 0 when absent, or 1 when present, and consisted of a history of physical attacks and fear-inducing behavior during the two weeks before hospital admission, absence of recent suicidal behavior, schizophrenia or manic diagnosis, male sex, and currently married or cohabiting. McNeil et al. evaluated this in a sample of 338 patients admitted to a university-based short-term inpatient unit over 20-months. After the study period, six clinical staff rated the charts and derived an ROC curve to define a score of ≤ 2 as low risk and > 2 as high risk of violence. They reported a positive predictive value (59.0%), negative predictive value (70.6%), sensitivity (57.2%), specificity (70.0%), total predictive value (65.4%), and likelihood ratio (1.97), indicating that the score might be of moderate value. The strength of these findings is the demonstration of an objective cutoff point classifying low and high risk with reasonable specificity and NPV; however, the tool is limited to the inpatient setting.

3.9. New tools

3.9.1. Aggressive Behavior Risk Assessment Tool (ABRAT)

The ABRAT [29] was originally developed to identify potentially violent medical–surgical patients. It consists of 10 violent behavior indicators in hospitalized patients, including a history of aggression, aggressive/threatening behavior, agitation, anxiety, confusion, shouting, mumbling, staring, and signs/symptoms of mania. Each item was evaluated on a dichotomous scale (absence = 0, presence = 1), with a summation that provides a score ranging from 0 to 10, and higher, indicating a higher risk of violence. Later, the original ABRAT was refined [30] to a checklist consisting of age < 85 years, history of physical aggression, confusion/cognitive impairment, anxiety, physical aggression/threatening, and threatening to leave. The checklist was evaluated in 615 newly admitted group home residents who were followed for 30 days. Aggressive events were recorded and included violence directed against staff or other residents. Using ROC analysis, a cut-off point of 4 provided a sensitivity and specificity of 55.6% and 94.2%, respectively, with PPV and NPV of 37.7 and 97.1, respectively [30].

Kim et al. [31] performed a prospective study in three EDS patients aged > 10 years. In this evaluation, the triage nurse collected the original

ABRAT data with six additional items that included a history of alcohol/drug abuse, history of mental illness, hallucinations, police custody, pacing, and significant intoxication. Overall, only 44% of patients had a calculated score. Recorded outcomes included all violent incidents in the ED for 60 days followed by a chart review. Of the 10,554 patients with ED, 127 had a violent event. Using multivariate logistic regression, they identified the most relevant predictors of violence as a history of aggression or mental illness, and the cause of ED visits as aggression or psychiatric evaluation, aggression threatening, agitation, staring, or confusion. Using a score cutoff of 1, it provided a sensitivity and specificity of 84.3% and 95.3%, respectively, with a PPV and NPV of 8% and 99.8%, respectively. The strength of this evaluation is the large numbers tested in the ED, with excellent sensitivity and specificity for future violence. Its limitations include evaluation in a single center, a limited number of violent patients ($n = 127$), and overall low compliance (44%) of participation, and it was performed during the covid-19 pandemic, which may affect generalizability [31].

3.9.2. The Emergent Documentation Aggression Rating Tool (EDART)

The EDART [31,32] is a quality improvement tool developed by three nurses to assess and document aggressive behaviors in the ED. It rates behaviors in six domains: (0) no aggression, (1) early indicator (crying, rocking, etc.), (2) signs of aggression (rudeness, glaring, etc.), (3) beginning of aggression (pacing, suicidal/homicidal ideation, etc.), (4) escalated aggression (yelling, level 1–3 behavior outside of room, etc.), and (5) danger to self and others (kicking, punching, etc.).

EDART was evaluated in 2018 and 2019 in the ED of a Midwestern suburban hospital. Its implementation resulted in a marked improvement in the documentation of violence, from 67.4% to 97.6%, and a numerical decrease in restraint use. The strength of this evaluation is the improvement of documentation and the potential for decreased restraint use, although it is limited by the lack of a validated scoring system and statistical rigor.

4. Discussion

Overall, we identified 10 assessment tools for the objective evaluation of potential violence in an acute setting. These studies vary in terms of patient population, methodology, objectives, and statistical rigor. Of these, eight were ED-based studies, with only three studies enrolling > 1000 patients. Of the ED-based studies, only three evaluated the predictive capability of a score and performed appropriate statistics. Finally, two studies provided clinical outcomes as a function of the score implementation.

The three largest studies included here involved the ABRAT, BVC, and EDART scores. First, the ABRAT expanded score was tested by Kim et al. [31], who enrolled 10,554 patients aged ≥ 10 years; of these, 627 were mental health patients and 127 were violent patients with ≥ 1 violent incident. The score was completed by ER triage nurses in three EDs, and the study was designed to evaluate the utility of the score in detecting violent episodes in conventional ED. Second, BVC [17] was evaluated in a single Australian ED in 2064 patients of which 35 committed violent acts, and the score was assessed by security officers (non-clinicians). The study was designed to determine the utility of the score when assessed by a healthcare security officer. Campbell et al. tested the EDART [32] scores in a conventional ED. Emergency room nurses evaluated 32,290 adult patients using the score. However, the purpose of the project was to evaluate the feasibility of using EDART and to capture the nurse-perspective of the tool. Even though the study had the largest number of patients in the ED setting, it was descriptive in nature, and the predictive capability of EDART was not reported.

Among the studies that reported 8% predictive capability of ABRAT [31], once again, the highest sensitivity and specificity (84.3% and 95.3%, respectively), with a capability to improve specificity to 98.9% and an AUC of 0.91. The BVC [15,16] demonstrated the highest specificity (99.4%), but its sensitivity was relatively poor (45.7%). BVC also

reports positive and negative predictive values at different cut-off scores, which may help to tailor its application. Lastly, the VSC was evaluated in hospitalized mental health patients, and a cut-off of >2 was found to have a sensitivity and specificity of 57.2% and 70%, respectively, for predicting violent episodes, with a likelihood ratio of 1.97 [15]. Among the three scores, VSC [28] was evaluated in the least number of patients and performed lower than ABRAT and BVC.

A total of 3 studies reported outcomes. First, the OVA [17] checklist uses the BVC [15] score to evaluate staff comfort in detecting violence and security activations in response to a violent episode. Implementation of the tool improved confidence in predicting, but not preventing, violence. The score lowered the number of emergent security activations (code grey), but the planned security activations were higher. No difference in the use of mechanical restraints was noted; however, the sample size was not sufficient to see a difference. Second, Legambi et al. [33] used the number of restraints and the time patients spent in restraints to assess the impact of using BARS. The results indicated a numerical trend for patients to spend less time on restraints, although the numbers were not adequate for statistical analysis. Finally, the EDART [32] measures nursing documentation and restraint use. The results showed a marked increase in documentation and a decrease in the number of restraints; however, the study was descriptive in nature and lacked statistical rigor.

The shortest scores, which may imply quick administration and, hence, of interest to ED providers, are BVC/OVA and BARS. BVC is comprised of only 6 items and required minimal training for its use. The BARS tool was also a short 7-item score, which was easily assessed by ER nurses and recommended by the BETA guidelines. None of the studies provided data on ease of use or time to administer.

5. Limitations

Our study has several limitations. By restricting the inclusion to studies conducted in an acute setting, we may have overlooked relevant factors that are applicable to the Emergency Department. Additionally, we only considered studies written in English, and it is possible that valuable scores developed in other languages may have been excluded. Furthermore, we excluded scores created to evaluate agitation in non-psychiatric patients, such as the elderly or dementia patients, which may be useful in the Emergency Department.

6. Recommendations and conclusions

In a formal context, it is imperative to predict aggression and agitation among psychiatric patients in busy emergency departments. The utilization of violence risk prediction scores is crucial for decision making, but the process can be complicated. The efficacy of the scoring system can be confidently evaluated by assessing its impact in an emergency setting. This involves examining the number of physical and chemical restraints employed prior to and after the use of the score, the number of violent incidents against staff or hospital property, and the number of security interventions. This evaluation helps to determine the effectiveness of the score and can improve patient care. Several violence risk prediction scores have been reported from data collected in acute settings, although the majority were evaluated solely through observation. The OVA/BVC checklist, which appears to be the most appropriate and applicable for implementation in emergency departments, is straightforward to implement and has been demonstrated to predict aggression with a specificity of 99.4%, with the added benefit of reducing the number of security call activations and increasing early intervention to prevent violent incidents. However, it should be noted that this checklist has only been tested in one Australian healthcare system. Therefore, it is crucial to externally validate the checklist and study the impact of utilizing OVA/BVC in emergency settings, while prospectively evaluating its accuracy in preventing aggression.

CRedit authorship contribution statement

Heba Mesbah: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Zubaid Rafique:** Writing – review & editing, Visualization, Methodology, Conceptualization. **Nidal Moukaddam:** Writing – review & editing, Visualization, Validation, Supervision. **William Frank Peacock:** Writing – review & editing, Supervision.

Declaration of competing interest

We declare no conflicts of interest.

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